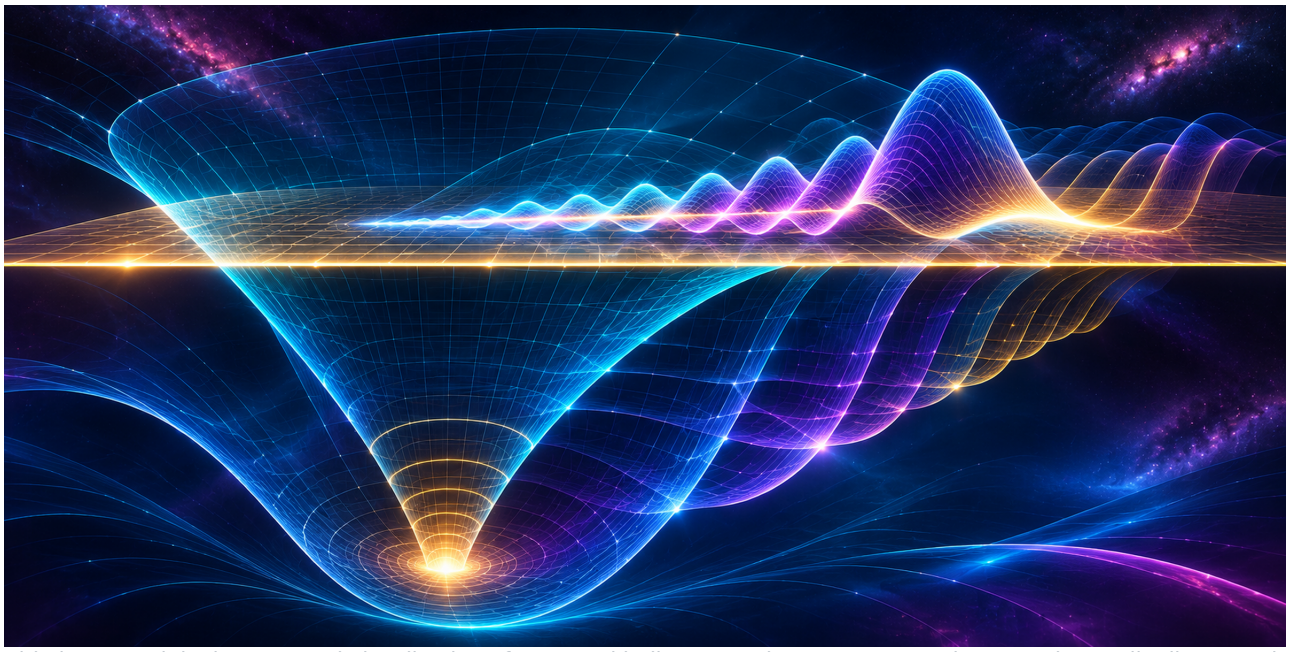
PACEVAL | SCIENTIFIC RESEARCH DEMONSTRATOR

Parameterized Bulk-Wave Evaluation Kernels with paceval

AdS5-inspired Fourier-mode waves, reusable analytical kernels, verified numerical convergence, real Multi-Graph-Batch execution and mathematically controlled formula revisions.



Title image: original conceptual visualization of a warped bulk wave. It is not a measured or experimentally discovered gravitational phenomenon. The numbered scientific figures use the actual computation outputs.

Verified scope: 28 reusable original-SDK mathematical graphs; 663,784 real batch results; 12 completed PDE time integrations; 2,779 independent 80-digit reference checks passed; CTest succeeded.

13 November 2026 | English research whitepaper | WP_044

1. Research use case and current scientific context

Computational physics and engineering workflows often solve a time-dependent PDE while repeatedly evaluating parameterized mathematical expressions over a spatial discretization. Interchangeable expressions are useful in reduced-order wave physics, spectral mode calculations, numerical-code verification, inverse modelling and repeated parameter surveys. The present study investigates whether the original paceval engine can act as an explicit and auditable mathematical execution layer inside such a workflow.

The original WP_044 proposal describes the bulk-wave acceleration $h_{tt} = -k^2 h + h_{zz} - 3 h_z/z$. This is consistent with the spatial operator appearing after Fourier separation in an AdS5 conformal-coordinate model. Its exact physical interpretation depends on gauge choice, polarization, boundary or junction conditions and the gravitational theory, none of which is inferred from a successful numerical run alone.

Scientific point of comparison

The Randall–Sundrum literature describes graviton radial modes in conformal coordinates with Bessel functions of order two. Numerical verification practice, including the NIST approach to the method of manufactured solutions, distinguishes correctness of an implementation from validation of its real-world physical model.

What this executable delivers	What it does not imply
Real reusable paceval expression handles	No observation of extra dimensions
Second-order finite-difference convergence on analytic modes	No real gravitational-wave detection
Source, RHS, energy, parameter and versioned expressions	No nonlinear 5D Einstein evolution
Deterministic finite-domain RK4 experiment	No proof of faster-than-light communication

2. Mathematical model and coordinate domain

The controlled problem is a linear Fourier-mode field on a fixed conformal bulk coordinate interval z in [1, 2]. The coefficient of h_z is inherited from the five-dimensional conformal metric. The additional potential q is a dimensionless numerical test parameter, not a derived physical particle mass. The separately versioned delta modifies that potential without overwriting the baseline graph.

$$h_{tt} = h_{zz} - \frac{3}{z} h_z - (k^2 + q + \delta)h + S(t, z)$$

Here k is the Fourier wave number along a suppressed spatial direction, t and z are dimensionless and S is a deliberately specified manufactured source. Boundary values are the exact solution at z=1 and z=2, so boundary work may cause the energy within this finite interval to change.

Admissible research input domain

Variable	Controlled range or value
z	1.0 to 2.0 (away from coordinate singularity z=0)
k	Positive, with sweeps from 0.55 to 2.75
q	Nonnegative numerical potential; sweep 0.10 to 1.20
delta	Zero in baseline; 0.55 in versioned homogeneous case
time horizon	0.6 dimensionless time units
spatial grids	25, 49, 97 nodes; refinement ratio 2
time integrator	Classical RK4 with timestep <= 0.30 spatial interval

The full metric and physical polarization constraints are not evolved by this scalar test harness. No ad hoc physical unit system is imposed on these normalized modes.

3. Independent exact modes and test forcing

The study uses three mathematically distinct radial profiles and a fourth versioned copy of the zero mode. Exact solutions and prescribed Dirichlet boundaries are constructed independently from the paceval acceleration graph.

Homogeneous polynomial zero mode

$$h_0(t,z) = a(1 + 0.08z^4)\cos(\omega t), \quad \omega^2 = k^2 + q + \delta$$

For the unmodified zero mode, q and delta are zero; in the revision, delta is 0.55 and the exact frequency is changed consistently. The radial operator annihilates both the constant and z-to-the-fourth components in the continuum equation.

Homogeneous oscillatory Bessel mode

$$h_B(t,z) = a\, z^2 J_2(mz)\cos(\omega t), \quad \omega^2 = k^2 + q + m^2$$

The order-two Bessel function comes from the exact homogeneous radial equation. C++ uses its standard-library implementation only to provide initial and Dirichlet data; the interior PDE acceleration is evaluated by paceval.

Forced manufactured mode

$$h_M(t,z) = az^2\sin(mz)\cos(\omega t)$$

$$S = a\cos(\omega t)[(k^2 + q - \omega^2 + m^2)z^2\sin(mz) + 4\sin(mz) - mz\cos(mz)]$$

The forcing is part of the actual paceval RHS, not a post-hoc adjustment. Choosing an exact field plus the corresponding source is a recognized numerical-code verification technique. This test does not claim a physically excited gravitational source.

4. Spatial operators, integration and energy diagnostics

Centered finite differences approximate the two radial derivatives. The paceval engine evaluates those differences, their warped combination and the complete right-hand side. C++ advances the coupled first-order state using a fixed-step fourth-order Runge–Kutta algorithm, imposing exact boundary values at each intermediate stage.

$$D_z h_i = \frac{h_{i+1} - h_{i-1}}{2\Delta z}, \qquad D_{zz} h_i = \frac{h_{i+1} - 2h_i + h_{i-1}}{(\Delta z)^2}$$

For smooth modes, centered differences have a second-order truncation error in space; with a time increment proportional to the mesh width, the fourth-order time-stepping error is smaller and the observed spatial convergence should approach order two.

Weighted energy and boundary exchange

$$E = \frac{1}{2}\int_1^2 z^{-3}(h_t^2 + h_z^2 + (k^2+q)h^2)\,dz$$

$$F = -z^{-3} h_t h_z, \qquad \Delta a = -\delta h$$

The weighted energy density and the flux proxy are both computed with persistent paceval expressions. Energy in the finite interval need not remain constant because Dirichlet boundaries move with the exact mode and because the manufactured test contains a source. Reported energy is a numerical diagnostic, not an assertion about conserved physical gravitational energy in a curved spacetime.

5. Actual original-SDK paceval execution

The C++ executable initializes the original paceval shared runtime, registers 28 expression handles once, and repeatedly evaluates subsets of these handles through the real Multi-Graph-Batch API. Every registered expression is actually exercised by the 7,623-point parameter survey. The dynamics use selected full RHS, weighted energy, error and flux graphs.

Mathematical function family	Actual execution owner
Warped first and second spatial differences	paceval: reusable graphs
Complete Fourier-mode acceleration, forced RHS	paceval: full expressions
Independent revision of RHS and potential	paceval: separate persistent graphs
Analytical source, exact manufactured field and sensitivities	paceval: reusable expressions
Energy density, flux, source power and residuals	paceval: reusable expressions
Stateful RK4 stepping, boundary assignment, state storage	C++: numerical orchestration
Bessel J2 exact boundary/initial reference	C++: standard-library special function
Independent 80-digit oracle	Python mpmath: validation only

The SDK interface returns a false status on successful multi-graph calls in this library revision; the implementation additionally checks the returned per-output error codes. Scalar-host and independently evaluated mathematical expressions are never counted as paceval graphs.

The detailed expression strings, actual API route and execution owner are recorded in examples/formula_registry.csv and results/math_execution_inventory.csv.

6. Reproduced executable outcomes

Quantity	Actual value
Original persistent paceval expression graphs	28
Successful original Multi-Graph-Batch calls	5,623
Successfully evaluated paceval outputs	663,784
Analytic or manufactured mode cases	4
Spatial resolutions for each case	25, 49, 97 grid points
Complete RK4 field simulations	12
Parameterized surface evaluation positions	7,623
C++ independent mathematical spot checks	1,260
Independent 80-decimal-digit values	2,779 passed
Maximum observed final L2 error	0.000309744
Real CMake/CTest original-SDK test	1 / 1 passed

The four cases are deterministic and deliberately constructed to allow an exact mathematical reference. A second-order convergent trajectory is not evidence that the physical AdS5 model is the unique or correct theory for any observed universe.

Source-data and output provenance

There are no measured laboratory, astronomical or gravitational-wave datasets in this package. The numerical dataset consists of dimensionless analytic initial and boundary values, exactly specified forcing and actual paceval C++ calculation outputs. The conceptual cover is not used as scientific data.

7. Numerical mode profiles and bulk-coordinate evolution

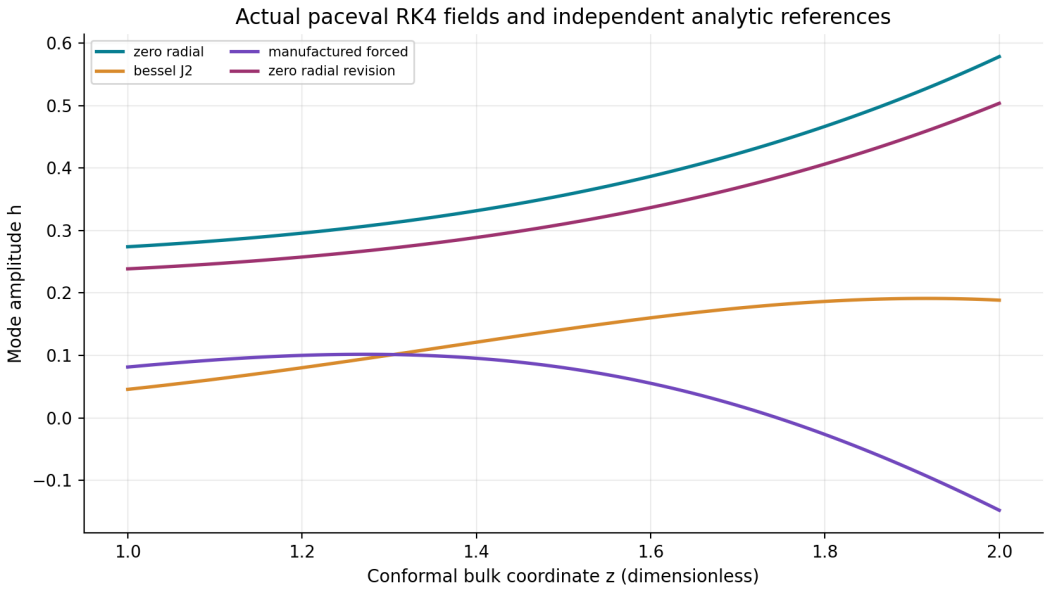


Figure 1. Four paceval-integrated final profiles (solid) alongside independent exact reference solutions (dashed).

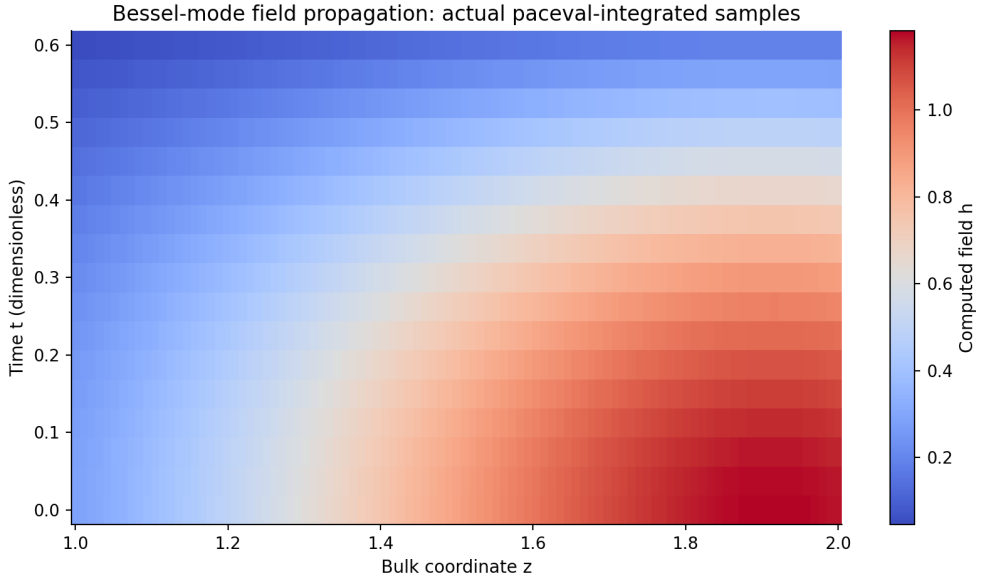


Figure 2. Actual Bessel-mode field values at logged time slices over the bulk coordinate.

8. Three-dimensional field and time-dependent error

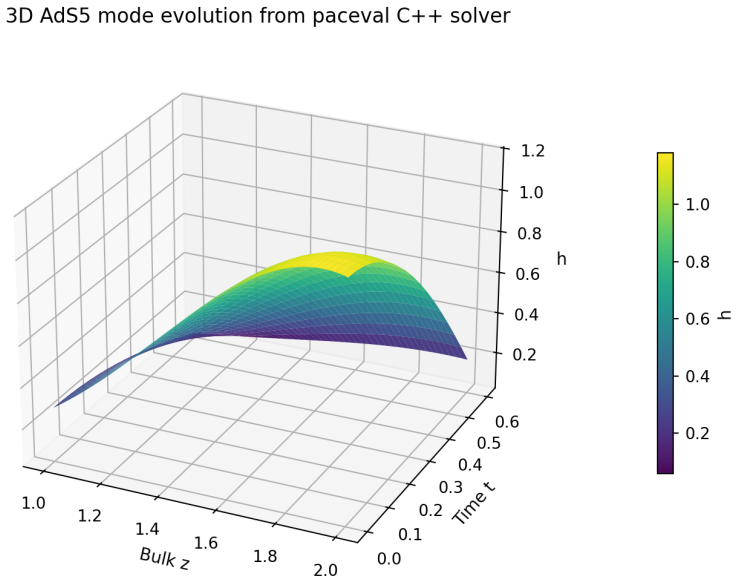


Figure 3. Genuine 3D visualization of the calculated Bessel-mode trajectory, not a simulated gravitational-wave detector.

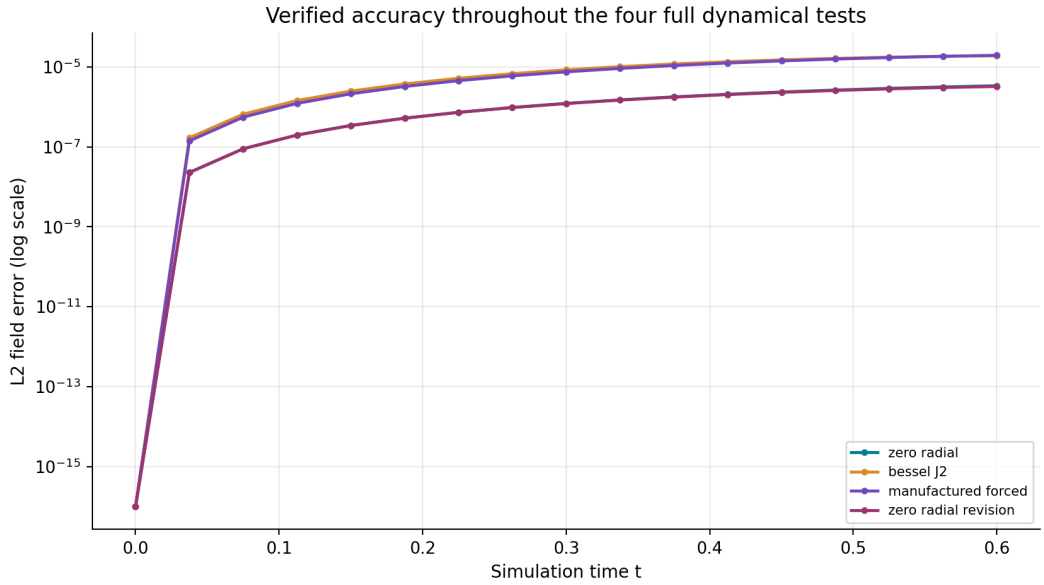


Figure 12. Field accuracy histories of all four test modes at 97 grid points.

9. Verification of spatial discretization order

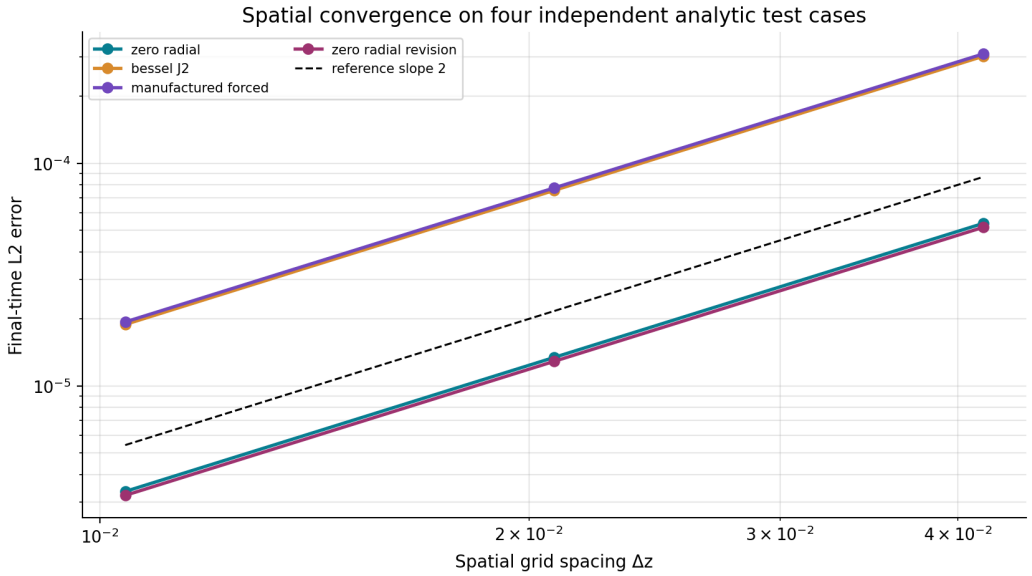


Figure 4. L2 error versus spatial step, showing systematic refinement towards the independently known analytic solutions.

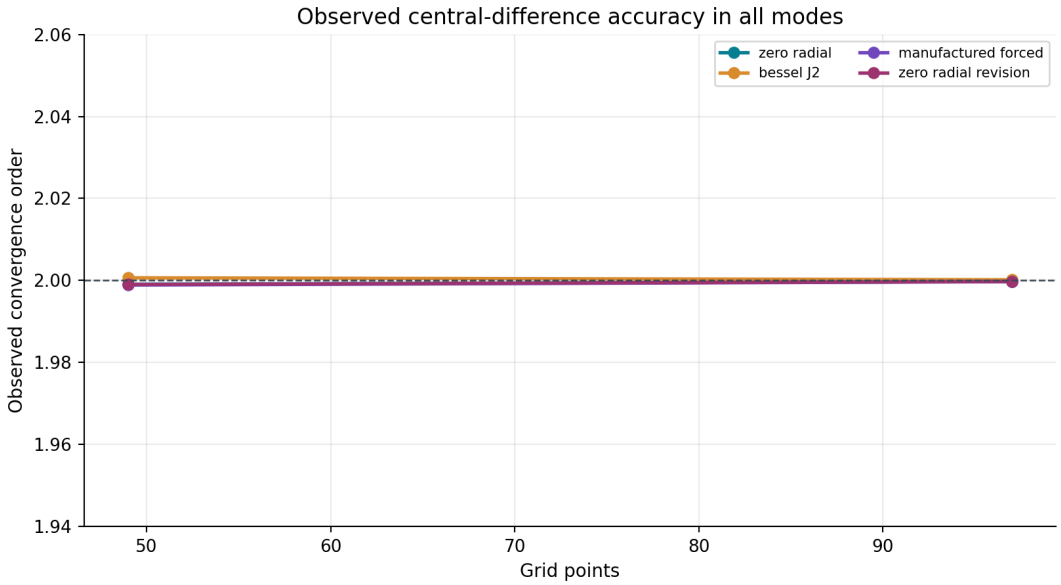


Figure 5. All four mode families approach the second-order accuracy of the centered spatial stencil.

10. Operator residual and manufactured forcing checks

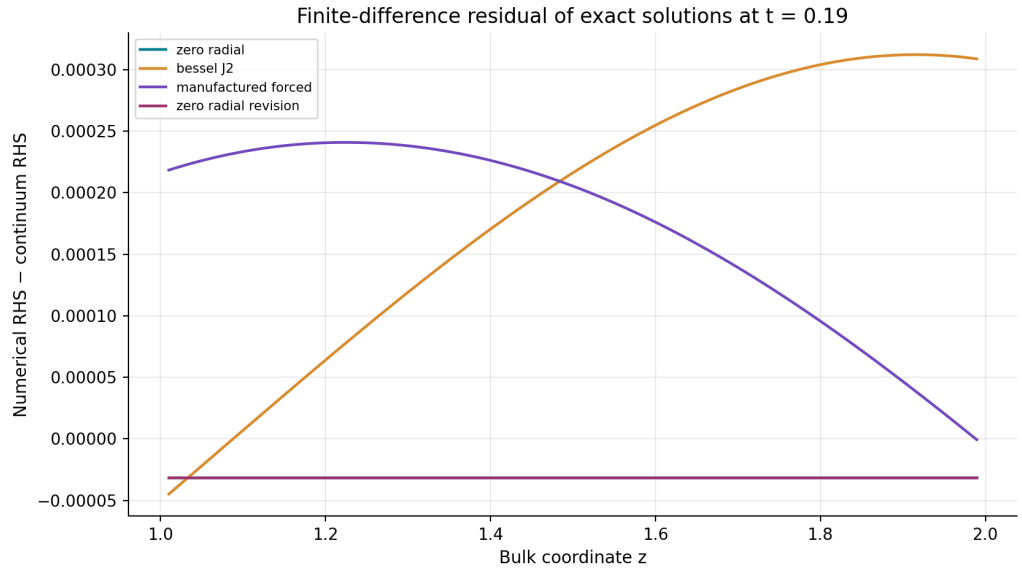


Figure 6. Finite-difference operator residual against each exact continuum wave acceleration.

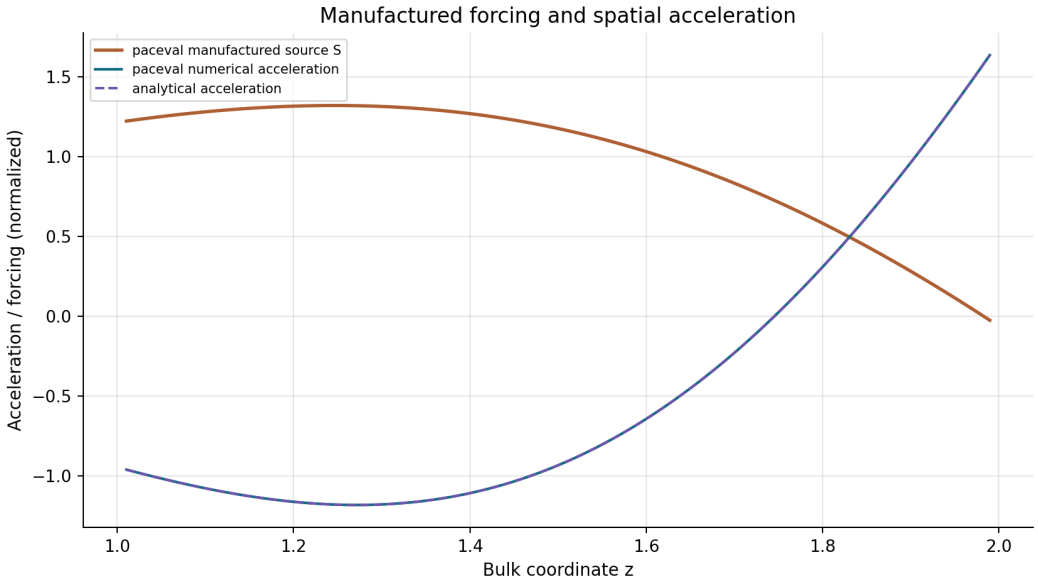


Figure 8. Real paceval source and numerical acceleration compared with the analytical manufactured acceleration.

11. Energy on a driven finite-domain interval

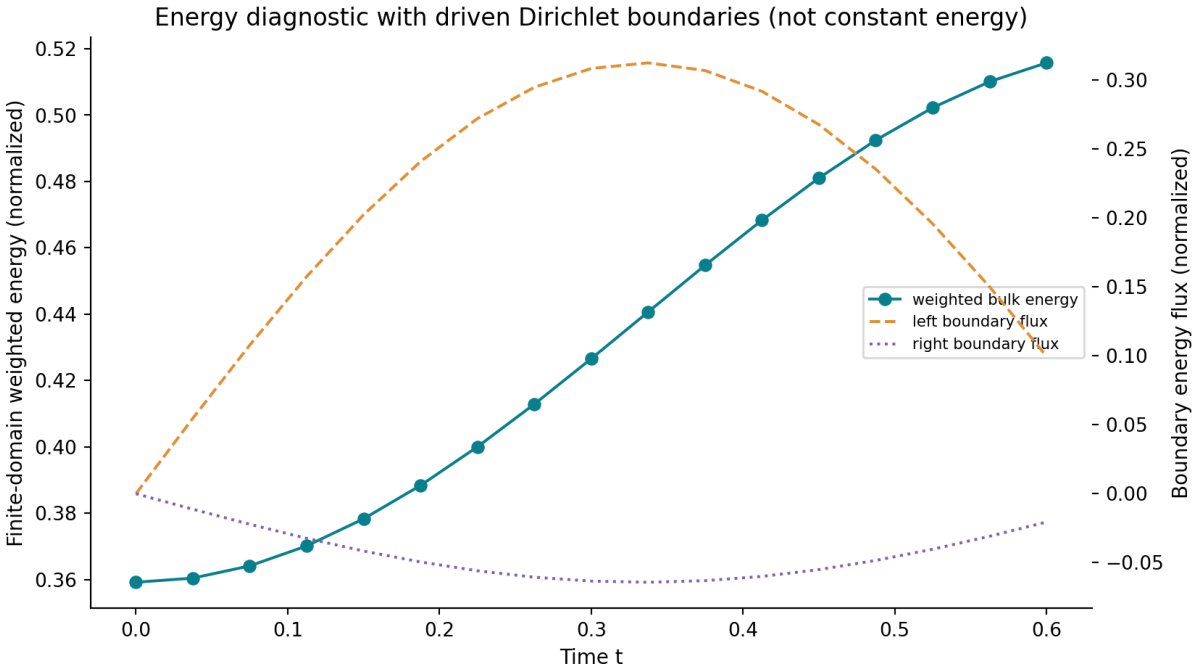


Figure 7. Weighted wave-field energy and estimated boundary flux, evaluated from paceval graph outputs.

The boundary data are time dependent. Consequently, a changing finite-domain energy is expected rather than evidence of a numerical instability or nonconservation of the full physical system. The present mesh-based flux is a diagnostic, not a demonstrated convergence of an independently derived continuum balance. An explicit high-order energy-flux audit remains a future experiment.

Practical numerical implication

The model supplies a useful architectural pattern: a physical or geometric kernel can be replaced or versioned without rewriting the C++ state integrator. The scientific usefulness of that architectural separation still depends on model appropriateness, stability and independent validation.

12. Bulk-wave parameter surfaces and numerical potential

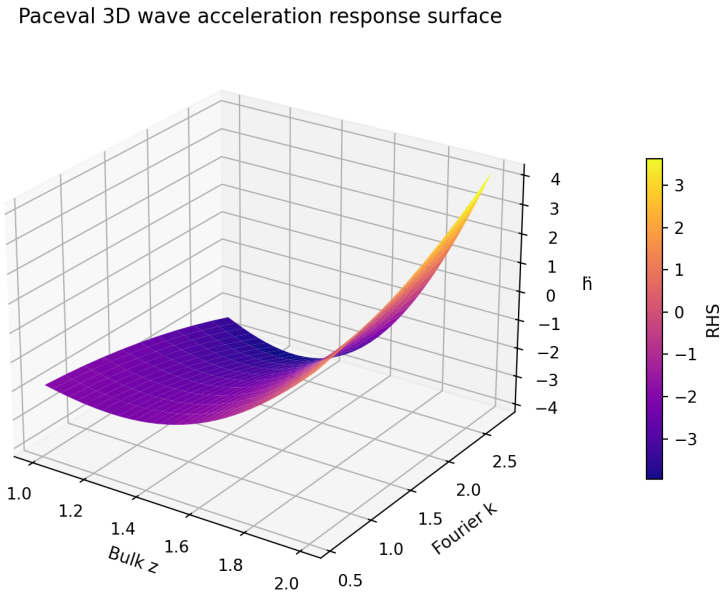


Figure 9. 3D acceleration of a fixed probe field (manufactured source disabled), generated by the real paceval graph sweep.

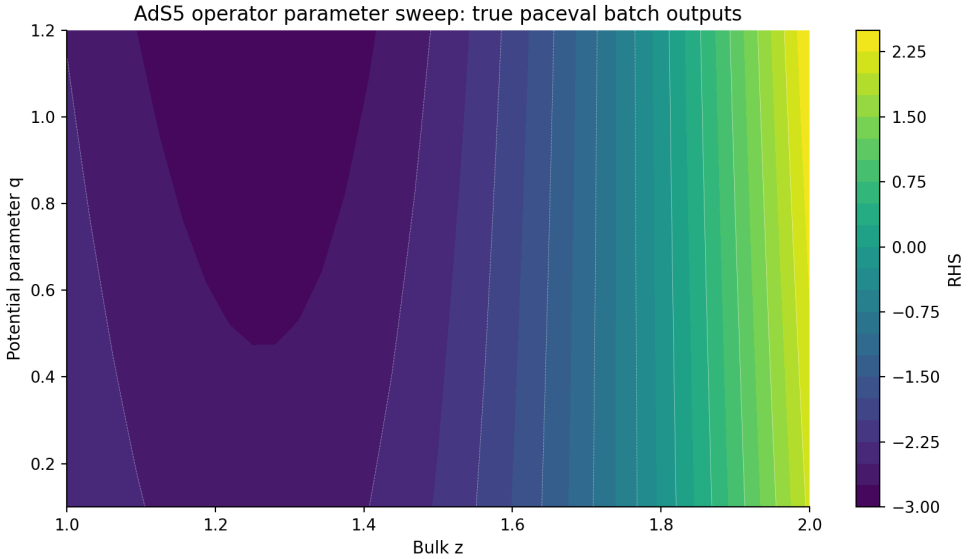


Figure 10. Unforced probe-field acceleration as the numerical potential q varies, with all other model inputs controlled.

13. Mathematical versioning and sensitivities

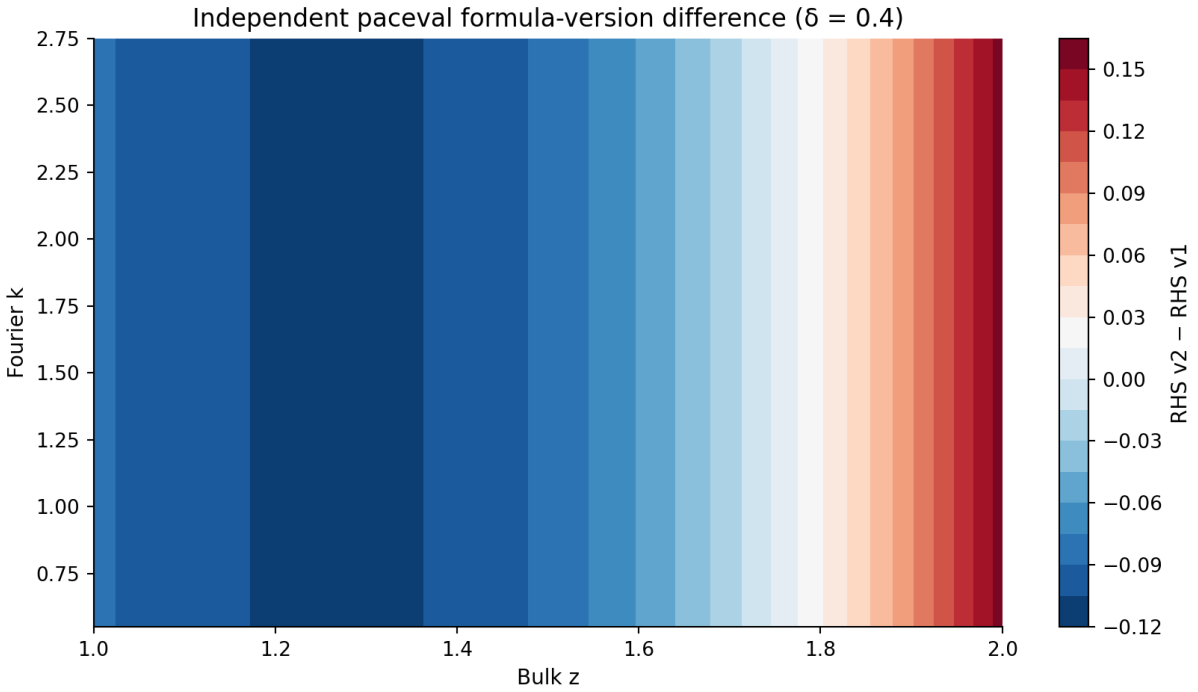


Figure 11. Independently registered formula-version difference, not a physical detection or new dispersion law.

The baseline RHS is retained unchanged. An additional registered expression subtracts delta times the field, so the revision is locally auditable. Explicit paceval sensitivities with respect to k, q and delta, together with versioned energy density and change functions, are included in the 28-graph registry and exercised by the parameter sweep.

Parameter gradients represent derivatives of the defined computational expressions. They cannot by themselves establish an independently meaningful physical interpretation for the tunable numerical potential.

14. Scientific interpretation, limitations and next steps

What has been established

The mathematical wave kernel is reusable and can be evaluated at many spatial points, time steps and parameter combinations with the actual paceval SDK. The independently specified analytic and manufactured solutions display the expected second-order spatial convergence. Multiple versioned expressions remain available, and core mathematical functions are executed by paceval rather than replaced with C++ numeric approximations.

What has not been established

This study does not solve nonlinear Einstein equations or gravitational backreaction. It does not derive junction conditions for a specific brane model, apply observable signal-transfer boundary conditions, infer a shortcut between spacetime events, or demonstrate gravitational propagation outside the light cone. The coordinate interval excludes the conformal boundary z=0; external exact Dirichlet boundaries are deliberately imposed. No physical measurement or spacecraft ephemeris is processed.

Scientific extension with falsifiable gates

Extension	Needed objective scientific check
Physical boundary conditions	Self-adjoint radial spectrum and flux consistency
Independent numerical solvers	Agreement against high-order and spectral discretizations
Physical five-dimensional gravity	Gauge and tensor constraints, junction conditions, backreaction
Actual bulk-to-brane signal analysis	Correct causal geometry, emitter-receiver boundary problem
Publication-oriented uncertainty	Numerical error bars, systematic variation and external review

15. Reproducibility, references and conclusion

The root README describes the build and reproduction commands. The included original paceval_main.h and paceval_main.cpp are linked to a separately obtained compatible original Linux shared runtime; its binary is intentionally excluded from this release. The 28 definitions are stored in examples/formula_registry.csv and the actual mathematical ownership in results/math_execution_inventory.csv.

Build: cmake -S src -B build -DCMAKE_BUILD_TYPE=Release -DPACEVAL_RUNTIME=/path/to/libpaceval_linux_sharedLIB.so

Run: cmake --build build -j; ctest --test-dir build --output-on-failure; ./build/bulk_wave_kernels --out . --n0 25 --time 0.6 --courant 0.30

Independent replay: python3 src/high_precision_reference.py; python3 src/verify_results.py; python3 src/make_figures.py; python3 src/make_whitepaper.py

Independent scientific sources

[1] The Randall–Sundrum Model: conformal coordinates and graviton modes. https://ncatlab.org/nlab/files/Gabella-RandallSundrumModel.pdf

[2] Durham University thesis: AdS conformal radial functions, Bessel order two. https://etheses.durham.ac.uk/id/eprint/2242/1/2242_252.pdf

[3] OSTI: AdS5 bulk wave mode equations. https://www.osti.gov/pages/servlets/purl/2579340

[4] NIST IR 8298, verification and validation methods for numerical simulation. https://nvlpubs.nist.gov/nistpubs/ir/2020/NIST.IR.8298.pdf

[5] Roache, Fundamentals of Verification and Validation; method of manufactured solutions. https://www.marin.nl/api/marin/downloads/get-download/6045/0/2704/block_files_hlee9l0uubi5p/8n76MTt2KkFhEwJLXf6U64eXsrI%3D/2009-FVV Book-Roache.pdf

[6] paceval C/C++ mathematical engine, original public SDK: https://github.com/paceval/paceval

[7] Original five-page WP_044 proposal (provided as part of this GitHub ZIP).

Conclusion

Verified numerical feasibility: 28 real paceval graphs, 663,784 validated batch outputs and independent reference checks demonstrate reusable, parameterized AdS5-inspired bulk-wave mathematics. The experiment does not establish a physical gravitational-wave shortcut or a fifth-dimensional observational signal.